

Extraneous Solutions and Real-World Applications - Practice

Algebra II Honors | Unit 2 | Lesson 6 Practice | TEK A2.6.D, A2.6.E

Objective: I can predict and verify whether a candidate solution to an absolute value equation is valid or extraneous, including all-extraneous cases.

Section 1 — Recall

1. Is $x = 3$ a valid or extraneous solution to $|3x - 2| = x + 4$?

$$|7| = 7, 3+4 = 7. \rightarrow \underline{\hspace{2cm}}$$

2. Is $x = -2$ a valid or extraneous solution to $|x - 1| = 3x + 8$?

$$|-3| = 3, 3(-2)+8 = 2. 3 \neq 2 \rightarrow \underline{\hspace{2cm}}$$

Section 2 — Apply

3. Before solving $|2x + 5| = 4x - 1$, state what x must satisfy for a solution to be valid. Then solve and check both candidates.

Requirement: _____

$x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$ Solution set: $\{\underline{\hspace{1cm}}\}$

4. Solve $|5x - 3| = 2x + 9$, checking both candidates.

$x = \underline{\hspace{1cm}}$ or $x = \underline{\hspace{1cm}}$ Solution set: $\{\underline{\hspace{1cm}}\}$ (both valid)

Section 3 — Challenge

5. Solve $|2x + 1| = -x - 10$. State the sign requirement first, then determine the full solution set.

Requirement: _____

Candidates: $x = \underline{\hspace{1cm}}$ and $x = \underline{\hspace{1cm}}$ - neither satisfies the requirement Solution set: _____

6. Solve $|3x - 4| = x + 2$, checking both candidates.

$x = \underline{\hspace{1cm}}$ or $x = \underline{\hspace{1cm}}$ Solution set: $\{\underline{\hspace{1cm}}\}$ (both valid)

Section 4 — Explain It

7. Explain why checking the sign of the right side at a candidate x -value is not just a helpful shortcut, but is mathematically GUARANTEED to correctly determine validity.
